

$$x > 0$$

$$xy' + y = y^2 \ln x \quad . a . 1$$

$$y' + \frac{1}{x} y = y^2 \frac{\ln x}{x}$$

$$z = y^{1-2} = y^{-1}$$

$$z' = -y^{-2} y'$$

1. a. 1

$$y^{-2} y' + \frac{1}{x} y^{-1} = \frac{\ln x}{x}$$

$$-z' + \frac{1}{x} z = \frac{\ln x}{x}$$

$$z' - \frac{1}{x} z = -\frac{\ln x}{x}$$

$$\mu(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\frac{1}{x} z' - \frac{1}{x^2} z = -\frac{\ln x}{x^2}$$

$$\left(\frac{1}{x} z\right)' = -\frac{\ln x}{x^2}$$

$$\int \frac{\ln x}{x^2} dx = -\frac{1}{x} \ln x + \int \frac{1}{x^2} dx = -\frac{1}{x} \ln x - \frac{1}{x}$$

$$\frac{1}{x} z = \frac{1}{x} \ln x + \frac{1}{x} + C$$

$$z = Cx + \ln x + 1$$

$$y = \frac{1}{z} = \frac{1}{Cx + \ln x + 1}$$

1. a. 1

$$y \equiv 0$$

$$\left(2x + \frac{y}{1+x^2y^2}\right) dx + \left(-2y + \frac{x}{1+x^2y^2}\right) dy = 0 \quad (1)$$

$$\left(2x + \frac{y}{1+x^2y^2}\right)_y = \frac{1+x^2y^2 - y \cdot 2x^2y}{(1+x^2y^2)^2} = \frac{1-x^2y^2}{(1+x^2y^2)^2}$$

$$\left(-2y + \frac{x}{1+x^2y^2}\right)_x = \frac{1+x^2y^2 - x(2xy^2)}{(1+x^2y^2)^2} = \frac{1-x^2y^2}{(1+x^2y^2)^2}$$

$$u = \int \left(2x + \frac{y}{1+x^2y^2}\right) dx = x^2 + \arctan xy + g(y)$$

$$u = \int \left(-2y + \frac{x}{1+x^2y^2}\right) dy = -y^2 + \arctan xy + h(x)$$

$$u = x^2 - y^2 + \arctan xy + C$$

$$x^2 - y^2 + \arctan xy = C$$

$$1^2 - 3 + \arctan \sqrt{3} = C$$

$$-2 + \frac{\pi}{3} = C$$

$$x^2 - y^2 + \arctan xy = -2 + \frac{\pi}{3}$$

$$y(1) = \sqrt{3}$$

$$x^2 y'' + x y' + 4y = 1 \quad .11.2$$

$$x = e^t \quad y(t) = y(e^t)$$

$$l(r) = r(r-1) + r + 4$$

$$l(r) = r^2 - r + r + 4 = r^2 + 4 \quad \pm 2i$$

$$\cos 2t \quad \sin 2t$$

$$y'' + 4y = 1$$

המשוואה החדשה

$$\cancel{p} = A$$

$$\cancel{p} = \cancel{p}'' = 0$$

$$4 \cdot A = 1 \quad A = \frac{1}{4}$$

$$y_p(t) = 1/4$$

$$y = C_1 \cos 2t + C_2 \sin 2t + 1/4$$

$$y = C_1 \cos 2 \ln x + C_2 \sin 2 \ln x + 1/4$$

$$y' = -C_1 \sin 2 \ln x \cdot \frac{2}{x} + C_2 \cos 2 \ln x \cdot \frac{2}{x}$$

$$0 = y(1) = C_1 + 1/4 \quad C_1 = -1/4$$

$$0 = y'(1) = 2C_2 \quad C_2 = 0$$

$$y = -1/4 \cos 2 \ln x + 1/4 \quad x > 0$$

$x > 0$ מניין טחטום ההתחלה פתרון
 $x_0 = 1$ והנצ"ר נ"ח מוצגת האפס וסוף
 $x < 0$ סתם מוצגת סתם סתם סתם
 $x > 0$ סתם מוצגת סתם סתם סתם

$$y'' - 4y = e^{-2x}$$

. n . 2

$$r^2 - 4 = (r-2)(r+2)$$

$$y_p(x) = x(Ae^{-2x}) = Ax e^{-2x}$$

$$y_p'(x) = Ae^{-2x} - 2Ax e^{-2x}$$

$$y_p''(x) = -2Ae^{-2x} + 4Ax e^{-2x}$$

$$-2Ae^{-2x} + 4Ax e^{-2x} - 4Ax e^{-2x} = e^{-2x}$$

$$-2Ae^{-2x} = e^{-2x}$$

$$A = -1/2$$

$$y_p = -1/2 x e^{-2x}$$

$$y'' - (2x^2 + 2)y' - 6xy = 0 \quad \text{Wronskian } -3$$

$$\sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n - 2x^2 \sum_{n=0}^{\infty} n a_n x^{n-1} - 2 \sum_{n=0}^{\infty} n a_n x^{n-1} - 6x \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n + \sum_{n=0}^{\infty} -2n a_n x^{n+1} + \sum_{n=0}^{\infty} -2n a_n x^{n-1} + \sum_{n=0}^{\infty} -6 a_n x^{n+1} = 0$$

$$\sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n + \sum_{n=1}^{\infty} -2(n-1) a_{n-1} x^n + \sum_{n=-1}^{\infty} -2(n+1) a_{n+1} x^n + \sum_{n=1}^{\infty} -6 a_{n-1} x^n = 0$$

$$2a_2 - 2a_1 + \sum_{n=1}^{\infty} \left[(n+1)(n+2) a_{n+2} - 2(n-1) a_{n-1} - 2(n+1) a_{n+1} - 6 a_{n-1} \right] x^n = 0$$

$$2a_2 - 2a_1 + \sum_{n=1}^{\infty} (n+1)(n+2) a_{n+2} - 2(n+2) a_{n-1} - 2(n+1) a_{n+1} = 0$$

$$a_0 = 1 \quad a_1 = 0$$

$$2a_2 - 2a_1 = 0 \Rightarrow a_2 = 0$$

$$a_{n+2} = 2 \frac{(n+2) a_{n-1} + (n+1) a_{n+1}}{(n+1)(n+2)} \quad n \geq 1$$

$$n=1 \quad a_3 = 2 \cdot \frac{3a_0 + 2a_2}{6} = 1$$

$$n=2 \quad a_4 = 2 \cdot \frac{4a_1 + 3a_3}{12} = \frac{1}{2}$$

$$n=3 \quad a_5 = 2 \cdot \frac{5a_2 + 4a_4}{20} = \frac{1}{5}$$

$$y(x) = 1 + x^3 + \frac{1}{2}x^4 + \frac{1}{5}x^5$$

$$y' + 4y = u_1(t) e^t$$

$$\mathcal{L}\{y'\}(s) + 4\mathcal{L}\{y\}(s) = e^{-s} \mathcal{L}\{e^{t+i \cdot 4}\}(s)$$

$$s\mathcal{L}\{y\}(s) - y(0) + 4\mathcal{L}\{y\}(s) = e^{-s} \mathcal{L}\{e \cdot e^t\}(s)$$

$$s\mathcal{L}\{y\}(s) - 1 + 4\mathcal{L}\{y\}(s) = e \cdot e^{-s} \frac{1}{s-1}$$

$$(s+4)\mathcal{L}\{y\}(s) = e \cdot e^{-s} \frac{1}{s-1} + 1$$

$$\mathcal{L}\{y\}(s) = e^{-s} \left(\frac{e}{(s-1)(s+4)} \right) + \frac{1}{s+4}$$

$$\frac{1}{(s-1)(s+4)} = \frac{1}{5} \left[\frac{1}{s-1} - \frac{1}{s+4} \right]$$

$$\mathcal{L}\{y\}(s) = e^{-s} \left[\frac{e}{5} \cdot \frac{1}{s-1} - \frac{e}{5} \frac{1}{s+4} \right] + \frac{1}{s+4}$$

$$y(t) = u_1(t) \left(\frac{e}{5} e^{t-1} - \frac{e}{5} e^{-4(t-1)} \right) + e^{-4t}$$

$$p(r) = \begin{vmatrix} 4-r & 1 \\ 2 & 3-r \end{vmatrix} = (4-r)(3-r) - 2 = 12 - 7r + r^2 - 2 = r^2 - 7r + 10 = (r-2)(r-5)$$

$$r=2 \quad \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{matrix} 2a+b=0 \\ b=-2a \end{matrix} \quad \begin{pmatrix} a \\ -2a \end{pmatrix} = a \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$r=5 \quad \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad a=b \quad \begin{pmatrix} a \\ a \end{pmatrix} = a \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$e^{2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}, e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\bar{x}_p(t) = C_1(t) e^{2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + C_2(t) e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{aligned} C_1'(t) e^{2t} + C_2'(t) e^{5t} &= 1 \\ -2C_1'(t) e^{2t} + C_2'(t) e^{5t} &= -1 \end{aligned}$$

$$C_1(t) = e^{-2t} - C_2'(t) e^{3t}$$

$$-2(e^{-2t} - C_2'(t) e^{3t}) e^{2t} + C_2'(t) e^{5t} = -1$$

$$-2 + 2C_2'(t) e^{5t} + C_2'(t) e^{5t} = -1$$

$$C_2'(t) = \frac{1}{3} e^{-5t}$$

$$C_1'(t) = e^{-2t} - \frac{1}{3} e^{-2t} = \frac{2}{3} e^{-2t}$$

$$C_1(t) = -\frac{1}{3} e^{-2t}$$

$$C_2(t) = -\frac{1}{15} e^{-5t}$$

$$\bar{x}_p(t) = -\frac{1}{3} e^{-2t} e^{2t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} - \frac{1}{15} e^{-5t} e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 3/5 \end{pmatrix}$$

~~$$\begin{pmatrix} -1/3 \\ 1/6 \end{pmatrix} - \begin{pmatrix} 1/15 \\ 1/15 \end{pmatrix} = \begin{pmatrix} -6/15 \\ 1/15 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 1/15 \end{pmatrix}$$~~